Connectors



File: connectors.kicad_sch

Channels



File: channels.kicad_sch

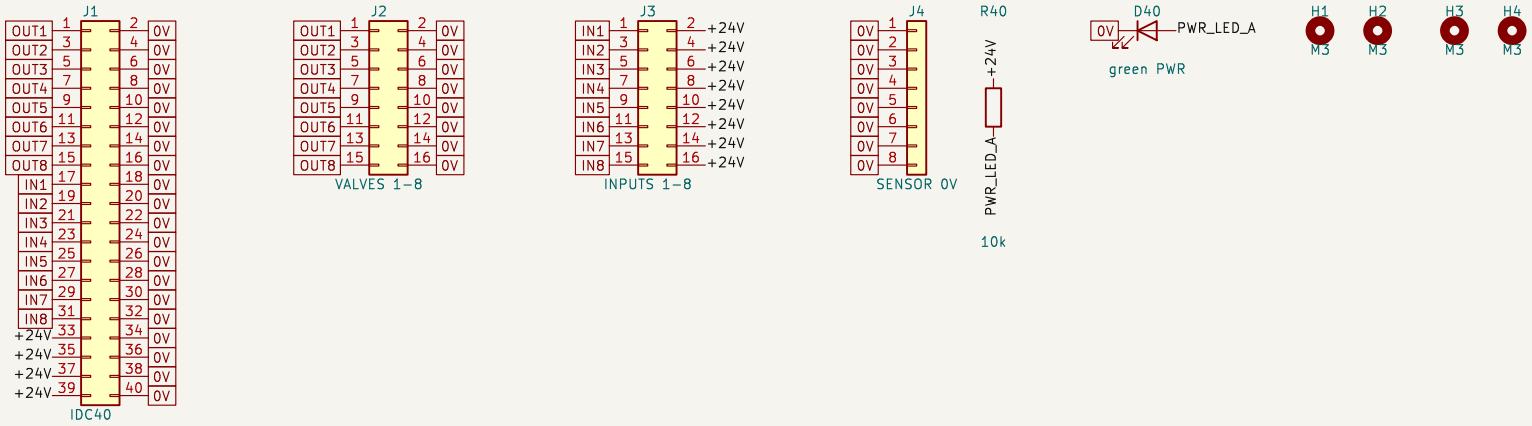
CEMS I/O breakout, rev 0.1 schematic (draft for review)
DIN-rail board: one 26-way latched IDC header (straight-through standard ribbon) to 24 screw terminals.
One design, two builds: OUT (valve bank A: flyback diodes fitted) and IN (digital inputs: no diodes).
Pinout, builds and parts: hardware/io_breakout/README.md. Parts marked CHECK need datasheet confirmation before layout.

Design spec: hardware/io_breakout/README.md
Generated from hardware/io_breakout/tools/design.py; do not edit by hand
EnviroDevelo

Sheet: /
File: cems_io_breakout.kicad_sch

Title: CEMS I/O breakout (IDC26 to terminals)

Size: A4	Date: 2026-10-30	**Rev: 0.1**
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Notes
J1: pin 1 marked on the silkscreen; keyed by the shroud notch, standard straight-through ribbon
J2: upper tier OUT1–8, lower tier 0V
J3: upper tier IN1–8, lower tier +24V

J4: 0 V for 3-wire sensors, one per input
R40: power LED, 2.2 mA at 24 V
H1: board 50 x 80 mm for a standard 50 x 80 mm DIN PCB holder (Adafruit 4435 for prototypes); CHECK hole positions

Generated from design.py
40-way latched IDC header, two-tier terminals, power LED, mounting
EnviroDevelo

Sheet: /Connectors/
File: connectors.kicad_sch

Title: CEMS I/O breakout: Connectors

Size: A4	Date: 2026-10-30	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/3



Notes
R1: 2.2 mA at 24 V, 48 mW
D1: flyback
RZ1: fitted = plain diode clamp; remove and fit Z for fast release
Z1: NOT FITTED. fast-release option. CHECK Vz + 0.7 V stays below the ISO2H823V2.5's own demagnetisation clamp
Z2: NOT FITTED.
Z3: NOT FITTED.

Z4: NOT FITTED.
Z5: NOT FITTED.
Z6: NOT FITTED.
Z7: NOT FITTED.
Z8: NOT FITTED.
R11: Input LED, 2.2 mA (the input itself draws about 2 mA)

Generated from design.py
Per output: LED, flyback diode, optional fast-release Zener. Per input: LED
EnviroDevelo

Sheet: /Channels/
File: channels.kicad_sch

Title: CEMS I/O breakout: Channels

Size: A3	Date: 2026-10-30	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/3